

- TP 2 : Explore AI PAAS Services
- Professor: Ouassim Karrakchou
- Level : 4th year

TP 2

AI PaaS Services

Objectives:

The goal of this TP is to introduce you to AI PaaS services in Microsoft Azure and experiment with them.

We will do the following tasks:

- Create an Azure Resource Group to organize the resources used during the TP.
- Create an Azure Storage Account where we can upload images to process using an AI service.
- Create an Azure AI Vision resource and call it using a REST API with Postman.
- Use Azure AI Vision to perform image analysis and text recognition (OCR).
- Deploy and interact with a Large Language Model (LLM) using Microsoft Foundry.
- Call the LLM through a REST API using Postman.
- Write simple Python programs to perform the REST calls previously done with Postman.

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher before midnight the day of the TP:
 - a. Awatif ROUIJEL awatifrouijel@gmail.com
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

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I- Create a Resource Group:

In this TP, we will explore several Microsoft Azure AI PaaS services and learn how to interact with them using REST API calls. We will mainly use Azure Blob Storage to store images, Azure Computer Vision to analyze images and extract text, and Microsoft Foundry to interact with a Large Language Model.

Many other AI services are available in Azure, such as Speech, Language, Document Intelligence, Content Understanding, Machine Learning, and many models available through Microsoft Foundry.

As a prerequisite for this TP, you need to have access to your Azure for Students account.

1. Access the Microsoft Azure Portal: <https://portal.azure.com/>
2. Log in using the account associated with your Azure for Students subscription.
3. In the search bar at the top of the Azure Portal, search for Resource groups.
4. Click on Create.
5. Select your Azure for Students subscription.
6. Set the name of the Resource Group to: **TP2-Cloud-Foundations**
7. Select an Azure region available to your subscription.
8. Click Review + create and then Create.

A Resource Group is a logical container used to organize Azure resources. All the resources that we will create during this TP will be placed inside this Resource Group. This will also make it easier to delete all the resources at the end of the TP.

Call the Professor to show him your Resource Group.

II- Create a Storage Account to store the images we will process:

The majority of the AI services we will explore today need some data to process. For computer vision applications, the input is usually an image.

There are different ways to send an image to an AI service: you can send the binary content of the image directly in the HTTP request or send the URL of an image accessible through the Internet. In many Cloud applications, the image is first uploaded to Cloud Storage and an AI service subsequently processes it.

We will therefore first create an Azure Storage Account and store our images inside an Azure Blob Storage container.

1. In the Azure Portal search bar, search for Storage accounts.
2. Click Create.
3. Configure the Storage Account as follows:
 - Subscription: Azure for Students
 - Resource group: TP2-Cloud-Foundations
 - Storage account name: choose a globally unique name.
 - Region: select an available region.
 - Performance: Standard

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- Redundancy: Locally-redundant storage (LRS)

The Storage Account name must be globally unique in Azure.

4. Keep the other parameters to their default values.
5. Click Review + create.
6. Click Create.
7. Wait for the deployment to finish and click Go to resource.

Create a Blob Container:

1. In your Storage Account, navigate to Data storage > Containers.
2. Click + Container.
3. Set the name to: **images**
4. Keep the anonymous access level private.
5. Click Create.

Upload an image:

1. Go on the Internet and save on your computer an image of your favorite meal (for example: burger, pizza, couscous, tajine, sushi).
2. Open your newly created images container.
3. Click Upload.
4. Select the image from your computer.
5. Click Upload.
6. Verify that your image appears inside the container.

Call the Professor to show him.

III- Generate a temporary URL to access your image:

Our Blob container is private. However, the Azure AI service must be able to download our image in order to analyze it.

Instead of making the image permanently public, we will generate a Shared Access Signature (SAS). A SAS gives temporary and restricted access to an Azure Storage resource.

1. Click on the image that you uploaded.
2. Search for the option allowing you to Generate SAS.
3. Allow the following permission: **Read**
4. Choose an expiration time sufficiently long to finish the TP.
5. Generate the SAS token and URL.
6. Copy the Blob SAS URL. It should look conceptually similar to:

`https://mystorage.blob.core.windows.net/images/myimage.jpg?sv=...`

- Open a new browser tab and paste the URL.
- Verify that you can see the image.

Important: the text after the ? contains security information allowing access to your Blob. Do not share this URL publicly.

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Call the Professor to show him the image through the SAS URL.

IV- Call the Azure Computer Vision API:

You are now ready to use Azure Computer Vision. Azure Computer Vision provides several computer vision functionalities, including image tagging, image captioning, object detection, people detection, and optical character recognition.

Create an Azure Computer Vision resource:

1. In the Azure Portal search bar, search for Computer Vision.
2. Create an Azure Computer Vision resource.
3. Configure it as follows:
 - Subscription: Azure for Students
 - Resource Group: TP2-Cloud-Foundations
 - Region: select an available region
 - Name: choose a unique name
 - Pricing tier: select the free tier if it is available to your subscription
4. Click Review + create.
5. Click Create.
6. Once deployment is complete, click Go to resource.

Retrieve your API credentials:

REST APIs generally require authentication so that only authorized users can access the service.

1. Inside your Azure AI Vision resource, navigate to the section containing Keys and Endpoint.
2. You should find KEY 1, KEY 2, and Endpoint.
3. Copy one API key and the Endpoint. Keep them somewhere because you will need them for the rest of the TP.

Important: an API key is a secret. You should never publish it or put it in a public GitHub repository.

Call the Professor to show him your resource and the location of Keys and Endpoint.

V- Image analysis using Postman:

We will now call Azure AI Vision using a REST API. A REST API allows applications to communicate with a service using standard HTTP requests such as GET and POST. In our case, we will send an HTTP POST request.

1. Download and install Postman if you do not already have it:
<https://www.postman.com/downloads/>
2. Create a new HTTP request and select POST.
3. Use the following URL:

ENDPOINT/computervision/imageanalysis:analyze?api-version=2024-02-01&features=tags

`https://myvisionresource.cognitiveservices.azure.com`

- Replace YOUR_API_KEY with the API key copied previously.

- ```
{
 "url": "BLOB_SAS_URL"
}
```

6. Click Send. You should receive a JSON response.

### Questions:

- Call the Professor to show him the results.**

Azure Computer Vision can also perform Optical Character Recognition (OCR). OCR consists of detecting and extracting textual information contained inside images.

- Create another Postman request. Use POST and the following URL:

Use the same HTTP headers:

- Use the following JSON body:

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```
{
 "url": "BLOB_SAS_URL"
}
```

Replace BLOB\_SAS\_URL with the URL of your image containing text. Click Send and study the returned JSON.

### Questions:

1. What text has been detected?
2. How are the different lines represented in the JSON response?
3. Does Azure return information about the location of the detected text?
4. Was all the text correctly recognized?

**Call the Professor to show him.**

## VII- Combine several AI Vision features:

So far, we requested one feature at a time. Azure AI Vision can request several visual features in the same API call.

Modify your request to:

**ENDPOINT/computervision/imageanalysis:analyze?api-version=2024-02-01&features=tags,caption**

Use your meal image and click Send.

### Questions:

1. What is the difference between a tag and a caption?
2. How many tags were returned?
3. What caption was generated by the model?
4. What confidence is associated with the caption?
5. Do you consider the description correct?

**Call the Professor to show him.**

## VIII- Interact with an LLM:

We will now use a Large Language Model (LLM). LLMs are generative AI models capable of processing natural language and generating text.

Examples of tasks that can be performed by LLMs include answering questions, summarizing text, translating, generating code, extracting information, and creating conversational assistants.

Azure provides access to generative AI models through Microsoft Foundry.

## Create a Microsoft Foundry project:

1. From the Azure Portal, search for Microsoft Foundry.
2. Open the Foundry service.
3. Create a new Foundry resource associated with TP2-Cloud-Foundations.
4. Select an available region.

5. Open the resulting Microsoft Foundry environment.

## Deploy a model:

1. Browse the available models.
2. Select a small chat/instruction model available to your Azure subscription like **gpt-5.4-nano**
3. Click Deploy.
5. Wait until deployment is complete.
6. Open the deployed model.
7. Locate the section allowing you to call the deployed model using an API.
8. Copy the REST endpoint and the API key.

**Call the Professor to show him the deployment.**

## IX- Call the LLM using Postman:

Real applications frequently interact directly with an LLM through an API. We will therefore communicate with our deployed model using Postman.

1. Create a new request and select POST.
2. Use the REST endpoint indicated in the Consume section of your model deployment.
3. Configure the authentication information by putting the API key in the **Authorization** section of Postman, with the **Bearer** type.
4. Add the following HTTP header:
  - **Content-Type: application/json**
5. In Body > raw > JSON, use the following:

```
{
 "messages": [
 {
 "role": "user",
 "content": "Explain cloud computing in three sentences."
 }
],
 "model": "gpt-5.4-nano",
 "max_completion_tokens": 4096,
 "reasoning_effort": "medium"
}
```

6. Click Send and study the JSON response.

### Questions:

1. Where is the text generated by the model located in the JSON response?
2. What other information is returned?
3. Can you find information concerning the number of tokens used?

**Call the Professor to show him.**

We will now investigate an important characteristic of REST calls to LLMs.

- My favorite programming language is Python. Remember this information.

3. Then create a new request containing only:

What is my favorite programming language?

- Create a request containing a conversation such as:

```
{
 "messages": [
 {
 "role": "user",
 "content": "My favorite programming language is Python."
 },
 {
 "role": "assistant",
 "content": "Understood."
 },
 {
 "role": "user",
 "content": "What is my favorite programming language?"
 }
]
}
```

### Questions:

- So far, you performed all the API calls manually using Postman. However, in a real Cloud application, these REST requests would generally be made by a program.

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The last part of the TP is therefore to write Python programs that perform the requests you previously tested using Postman. This part is intentionally less guided: you have to search for the solution on your own.

You can use the Python requests library:

```
import requests
```

**Call the Professor to show him the working Python programs.**

## XII- Delete your Azure resources:

Cloud services may continue consuming your Azure credits even when you are no longer actively using them. At the end of the TP, you must therefore delete the resources you created.

Because we created all our resources inside the same Resource Group, Azure makes this easy.

1. Open the Azure Portal.
2. Search for Resource groups.
3. Open:

TP2-Cloud-Foundations

4. Verify that it contains the resources created during the TP.
5. Click Delete resource group.
6. Follow the confirmation procedure requested by Azure.
7. Verify that the Resource Group has disappeared.

Deleting the Resource Group deletes the resources contained inside it, including the Storage Account and AI resources created for the TP.

**Call the Professor to show him that the resources have been deleted.**

### XIII- Assignment:

Based on what you learned in the TP explain the different steps needed to use Azure AI PAAS services using REST API calls and show an example of usage for images and texts. Use screenshots from the Azure portal and from Postman. Make sure you add the python code you used to query the API.

**When you finish, make sure you delete your resource group as per part XII.**